

# Nuclear Fuel Performance

NE-533  
Spring 2026

# MECHANICS

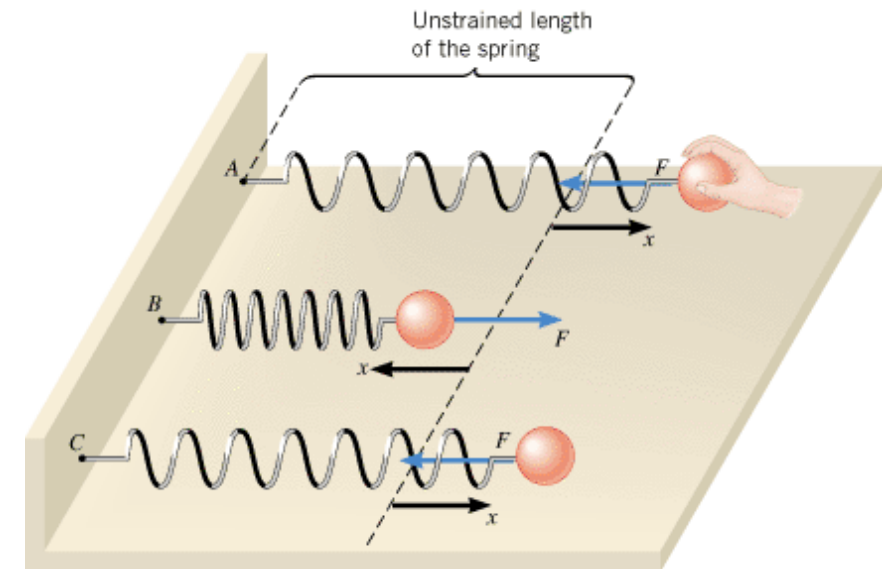
# Solid mechanics

- When a load is applied to a body, it changes shape and perhaps size
- Motions throughout a body are called **displacements**
  - $\mathbf{u}(\mathbf{r}, t)$
- Rigid body displacements do not change the shape and/or size
  - the rigid body is translated
- Changes in shape and/or size are call **deformations**
- The objective of **Solid Mechanics** is to relate loads (applied force) to deformation



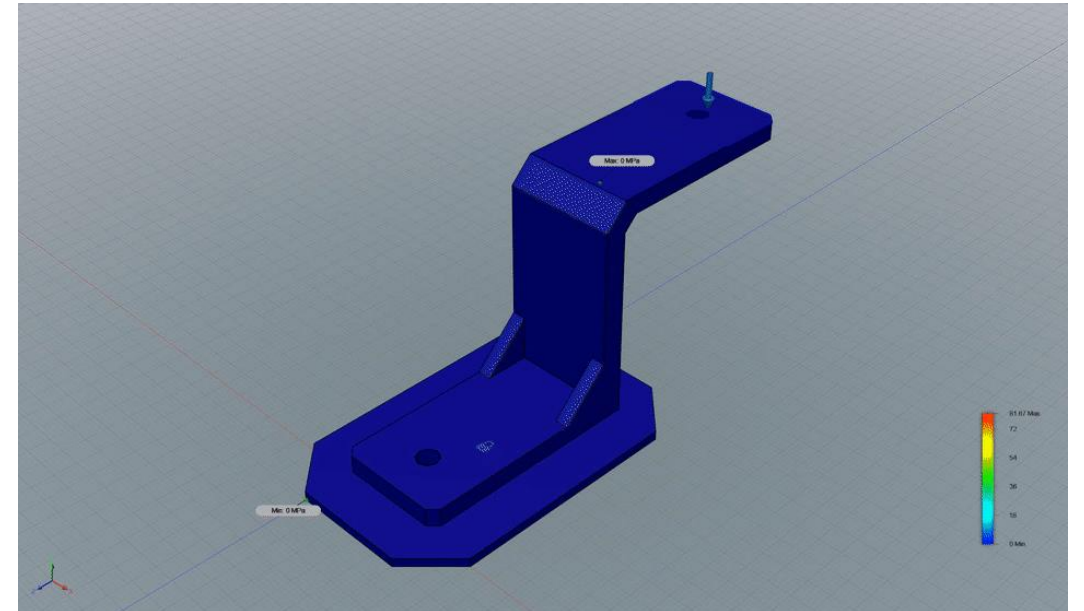
# Ideal springs

- It can be instructive to view solids as a spring
- When we apply some force  $F$ , we get some displacement  $x$ 
  - $F = kx$
- When the spring is displaced by  $x$ , there is force that responds in the opposite direction equal to  $kx$
- Due to the displacement, there is a stored energy  $E = \frac{1}{2} k x^2$



# Observed deformation due to a force

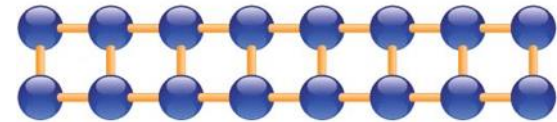
- Solid mechanics is similar to the behavior of an ideal spring but throughout a body
- An applied load results results in deformation.
- The internal **strain** is like the displacements  $x$
- The internal **stress** is like the internal force  $F$



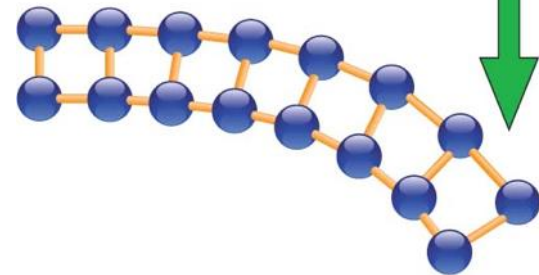
# Elasticity

- Elasticity is the ability of a body to resist a distorting influence and to return to its original size and shape when that influence or force is removed
- In elastic deformation, we are stretching the atomic bonds
- The more we stretch the bonds, the more force it takes to stretch
- When we release the load, the atoms spring back into their lattice sites

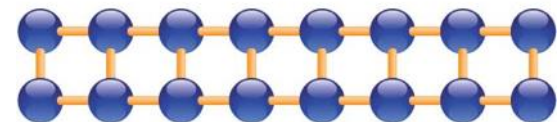
1. Original form



2. Force applied...

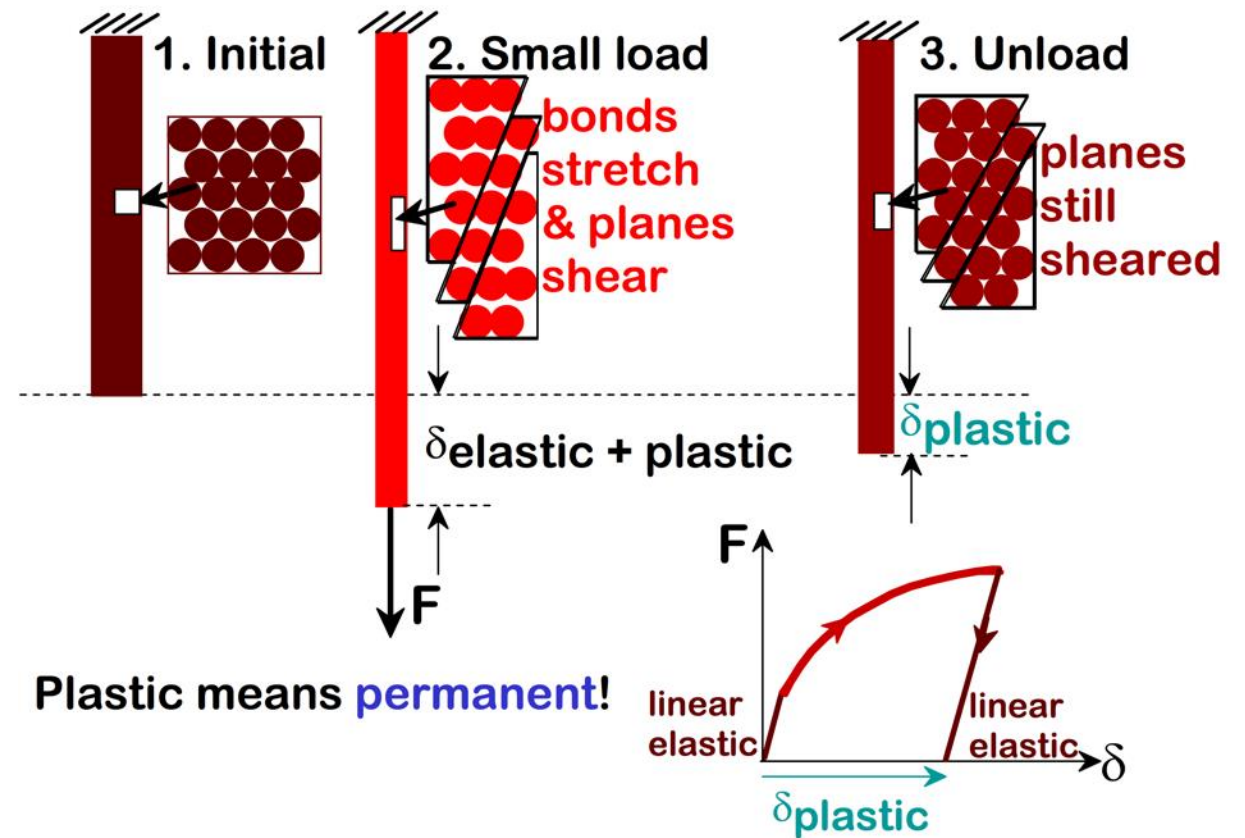


3. ...return to original form.



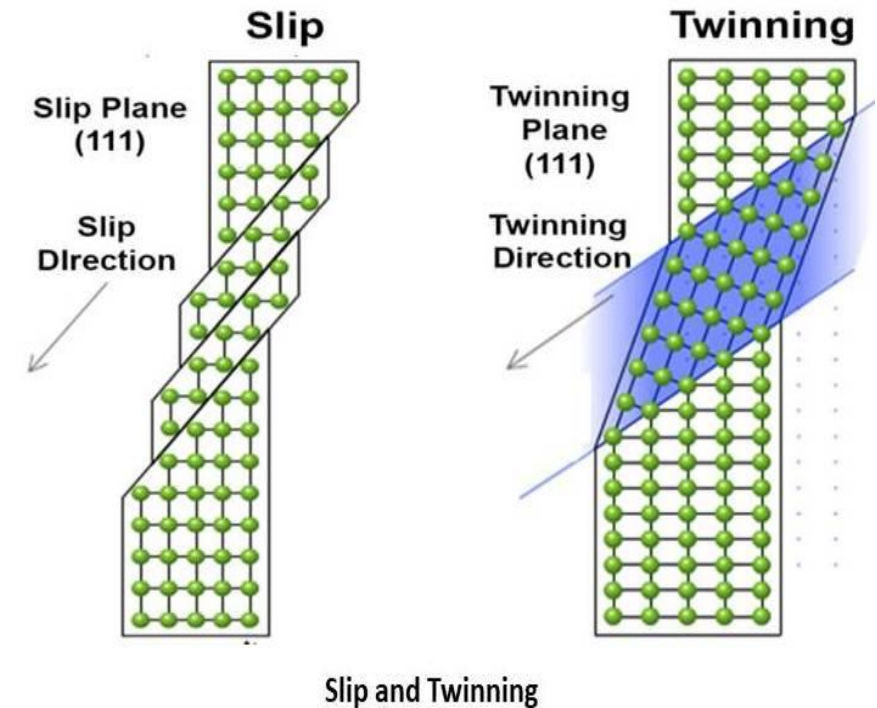
# Plasticity

- Plasticity is the ability of a solid material to undergo permanent deformation, a non-reversible change of shape in response to applied forces
- Plastic deformation is observed in most materials
- The transition from elastic behavior to plastic behavior is known as yielding



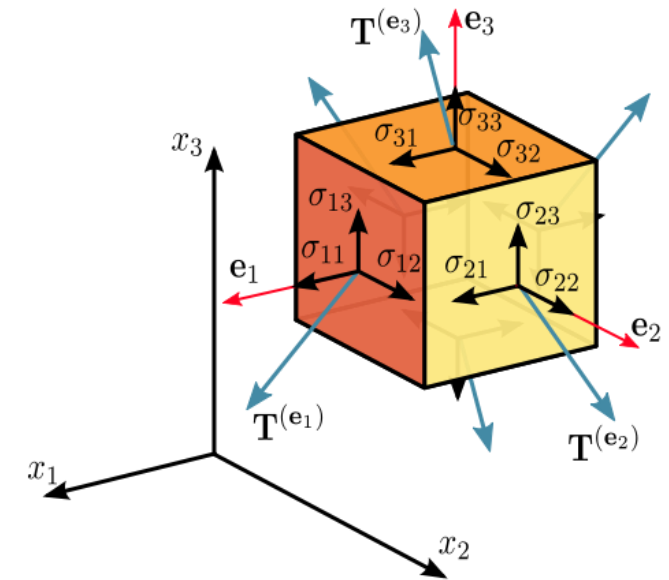
# Plasticity

- Plasticity is typically caused by two modes of deformation in the crystal lattice: slip and twinning
  - Slip is a shear deformation which moves the atoms through many interatomic distances relative to their initial positions
  - Twinning is the plastic deformation which takes place along two planes due to an applied force
- Ductility (total elongation), Yield Strength, etc. are dependent upon temperature and composition
- Plasticity typically increases at higher temperature



# Stress

- Stress is a physical quantity that expresses the internal forces that neighboring particles of a continuous material exert on each other
- Stress is a force per unit area with SI units of  $\text{Pa} = \text{N/m}^2$
- The stress is a 2<sup>nd</sup> order tensor (a 3 by 3 matrix): Cauchy stress tensor
- $\sigma_{ij} = F_{ij}/A_i$ 
  - i is the face the force is applied and j is the direction it is applied
- Sigmas are normal stress components, taus are shear stress components



$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \equiv \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix} \equiv \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

# Stress as a response

- The stress in a material is the RESPONSE to an applied load (force) or an applied displacement
- Uniaxial tension or compression
  - Only one non-zero stress:  $\sigma_{ii}$  ( $\sigma_{11}$ ,  $\sigma_{22}$ ,  $\sigma_{33}$ )
  - Tension means positive stress, compression negative
  - Examples: Cables, tension tests
- Pure shear
  - Only one non-zero stress: ( $\sigma_{12}$ ,  $\sigma_{13}$ ,  $\sigma_{23}$ )
  - Examples: drive shaft
- Biaxial tension/compression
  - Two non-zero stress (e.g.  $\sigma_{11} = 1$ ,  $\sigma_{22} = 2$ )
  - Examples: Pressure cylinder or vessel
- Hydrostatic compression (pressure)
  - $\sigma_{11} = \sigma_{22} = \sigma_{33} = -p$
  - Anything underwater

